CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 21: Surface Area, Volume and Capacity

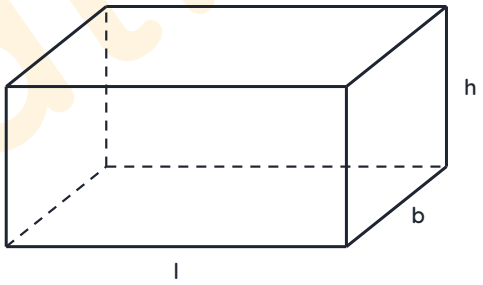
Topic-wise short notes with solved examples

Surface area and volume of the cuboid, cube and cylinder, the units of volume and capacity, and problems on boxes, tanks, walls and pipes.

1 Cuboid

- A **cuboid** has length l, breadth b and height h. It has 6 rectangular faces; opposite faces are equal.

Quantity	Formula
Volume	$l \times b \times h$
Total surface area	$2(lb + bh + hl)$
Lateral surface area (area of the four walls)	$2(l + b) \times h$
Diagonal	$\sqrt{l^2 + b^2 + h^2}$



Cuboid with length l, breadth b and height h

- **Lateral surface area** = perimeter of the base $\times$ height.
- A box open at the top has surface area $lb + 2(l + b)h$.

Example 1.1 A cuboid is 15 cm long, 10 cm wide and 8 cm high. Find its volume and total surface area.

SOLUTION

Volume $= 1200$. Surface area $= 2(150 + 80 + 120)$.

1200 cu cm; 700 sq cm

Example 1.2 A room is 6 m long, 5 m wide and 3.5 m high. Find the cost of whitewashing its four walls at ₹40 per sq m.

SOLUTION

Area of the walls $= 2(6+5) \times 3.5 = 77$ sq m

₹3080

Example 1.3 The volume of a cuboid is 720 cu cm, its length is 12 cm and its breadth is 10 cm. Find its height.

SOLUTION

$720 \div (12 \times 10)$

6 cm

2 Cube

- A **cube** is a cuboid with all edges equal to a.

Quantity	Formula
Volume	a^3
Total surface area	$6a^2$
Lateral surface area	$4a^2$
Diagonal	$a\sqrt{3}$

- If the edge is doubled, the surface area becomes 4 times and the volume 8 times.

Example 2.1 The total surface area of a cube is 486 sq cm. Find its edge and volume.

SOLUTION

$6a^2 = 486$, so $a = 9$ cm

9 cm; 729 cu cm

Example 2.2 Three cubes of edge 6 cm each are joined end to end. Find the surface area of the cuboid formed.

SOLUTION

The cuboid is 18 cm by 6 cm by 6 cm: $2(108 + 36 + 108)$.

504 sq cm

Example 2.3 How many cubes of edge 5 cm can be cut from a cube of edge 20 cm?

SOLUTION

$$\frac{20^3}{5^3}$$

64

3 Units of volume and capacity

- **Volume** is the space a solid occupies. **Capacity** is the quantity a container can hold.

Conversion	Value
1 cu cm	1 mL
1 litre	1000 cu cm
1 cu m	$1{,}000{,}000$ cu cm $=$ 1000 litres $=$ 1 kilolitre
1 cu cm	1000 cu mm

Example 3.1 A tank is 2.5 m long, 2 m wide and 1.6 m deep. How many litres of water does it hold?

SOLUTION

Volume $=$ 8 cu m

8000 litres

Example 3.2 Water is poured into a tank 80 cm long and 50 cm wide to a depth of 30 cm. Find the volume of water in litres.

SOLUTION

$80 \times 50 \times 30 = 120000$ cu cm

120 litres

4 Walls, boxes and bricks

- Number of bricks $=$ volume of the wall $\div$ volume of one brick (leave out the space taken by mortar if given).
- For a closed box of thickness t: each internal dimension $=$ external dimension $-2t$. Volume of material $=$ external volume $-$ internal volume.

Example 4.1 How many bricks, each 20 cm by 10 cm by 7.5 cm, are needed for a wall 6 m long, 3 m high and 20 cm thick?

SOLUTION

Wall: $600 \times 300 \times 20 = 3{,}600{,}000$ cu cm. Brick: 1500 cu cm.

2400

Example 4.2 A closed wooden box measures 40 cm by 30 cm by 20 cm on the outside and the wood is 1 cm thick. Find the volume of the wood.

SOLUTION

Inside: 38 cm by 28 cm by 18 cm $= 19152$ cu cm. Outside: 24000 cu cm.

4848 cu cm

5 Cylinder

- A **right circular cylinder** has two equal circular ends of radius r and height h.

Quantity	Formula
Volume	$\pi r^2 h$
Curved surface area	$2\pi r h$
Total surface area	$2\pi r(r+h)$
Area of each end	πr^2

- The curved surface, when cut and opened flat, is a rectangle with sides $2\pi r$ and h.
- **Hollow cylinder** (pipe) with outer radius R and inner radius r: volume of material $= \pi(R^2 - r^2)h$.
- Take $\pi = \dfrac{22}{7}$ unless told otherwise.

Example 5.1 Find the volume, curved surface area and total surface area of a cylinder of radius 7 cm and height 10 cm.

SOLUTION

Volume $= \dfrac{22}{7} \times 49 \times 10$. Curved surface $= 2 \times \dfrac{22}{7} \times 7 \times 10$. Total $= 2 \times \dfrac{22}{7} \times 7 \times 17$.

1540 cu cm; 440 sq cm; 748 sq cm

Example 5.2 The curved surface area of a cylinder is 1320 sq cm and its height is 15 cm. Find its radius and volume.

SOLUTION

$2 \times \dfrac{22}{7} \times r \times 15 = 1320$ gives $r = 14$ cm. Volume $= \dfrac{22}{7} \times 196 \times 15$.

14 cm; 9240 cu cm

Example 5.3 A cylindrical tank has diameter 2.8 m and height 3 m. Find its capacity in litres.

SOLUTION

$\dfrac{22}{7} \times 1.4^2 \times 3 = 18.48$ cu m

18,480 litres

Example 5.4 A metal pipe is 70 cm long with outer radius 5 cm and inner radius 4 cm. Find the volume of metal.

SOLUTION

$\frac{22}{7} \times (25 - 16) \times 70$

1980 cu cm

Example 5.5 A rectangular sheet 44 cm by 20 cm is rolled along its length to form a cylinder. Find its volume.

SOLUTION

Circumference $= 44$ cm gives $r = 7$ cm; height 20 cm.

3080 cu cm

6 Melting and recasting

- When a solid is melted and recast, or a liquid is poured from one container to another, the **volume stays the same**.

Example 6.1 A solid metal cuboid 22 cm by 14 cm by 7 cm is melted and recast into a cylinder of radius 7 cm. Find the height of the cylinder.

SOLUTION

$\frac{22}{7} \times 49 \times h = 2156$

14 cm

7 Common mistakes

- Confusing total surface area with lateral (or curved) surface area.
- Using the diameter in place of the radius.
- Writing 1 cu m $= 100$ litres. It is 1000 litres.
- Subtracting the thickness only once for a closed box.
- Mixing centimetres with metres in the same calculation.
- Giving area units (sq cm) for a volume or volume units for an area.